

Insider Access Anomaly Detection Report

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Project Overview

This case study investigates user access activity to identify anomalous behavior that may indicate insider threats, credential misuse, or unauthorized access attempts. Security analytics techniques were applied to examine user activity patterns, detect deviations from expected behavior, and prioritize suspicious events for further investigation.

Objective

The objective of this investigation was to analyze user access records, identify abnormal access patterns, and demonstrate how data-driven security analytics can support insider threat detection within enterprise environments.

Methodology

User access data was imported into a Jupyter Notebook and processed using Python. The dataset was cleaned and analyzed to identify unusual login behavior, abnormal access frequencies, and deviations from expected user activity. Statistical analysis techniques were used to highlight potentially suspicious accounts for further review.

Tools and Technologies

Python

Jupyter Notebook

Pandas

NumPy

Matplotlib

Key Findings

- Processed and analyzed user access activity.
- Identified anomalous user behavior based on access frequency and activity patterns.
- Demonstrated how security analytics can reduce investigation time by highlighting high-risk events.
- Produced a repeatable workflow suitable for insider threat investigations.

Security Relevance

User Behavior Analytics (UBA) is a critical capability in modern Security Operations Centers (SOCs). Detecting abnormal access behavior helps identify compromised accounts, privilege misuse, and insider threats before they result in significant business impact.

Skills Demonstrated

Security Analytics

Threat Hunting

Insider Threat Detection

User Behavior Analytics

Python Programming

Data Analysis

Investigation Methodology

Dataset Notice

The original dataset used during this investigation is not included in this repository due to course licensing and data-sharing restrictions. This repository includes only the implementation methodology, documentation, and summarized findings.

Conclusion

This project demonstrates a practical workflow for identifying anomalous user access behavior using security analytics. While statistical analysis alone cannot confirm malicious intent, it provides a valuable mechanism for prioritizing suspicious activity and supporting enterprise incident investigations.